QR Decomposition with Gram-Schmidt

Gram-Schmidt Process: Introduction

- The QR decomposition (also called the QR factorization) of a matrix is a decomposition of the matrix into an orthogonal matrix and a triangular matrix.
- A QR decomposition of a real square matrix A is a decomposition of A as:

$$A = QR,$$

  where Q is an orthogonal matrix (i.e. $Q^T Q = I$) and R is an upper triangular matrix.
- If A is nonsingular, then this factorization is unique.
- There are several methods for actually computing the QR decomposition. One such method is the Gram-Schmidt process.

Gram-Schmidt Process: Procedure

Consider the Gram-Schmidt procedure, with the vectors to be considered in the process as columns of the matrix A. That is,

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & | & \dots & | & \mathbf{a}_n \end{bmatrix}$$

Then, we compute the orthogonal vectors $\mathbf{u}_k$ and orthonormal vectors $\mathbf{e}_k$:

$$\mathbf{u}_1 = \mathbf{a}_1, \qquad \mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$$

$$\mathbf{u}_2 = \mathbf{a}_2 - (\mathbf{a}_2 \cdot \mathbf{e}_1)\mathbf{e}_1, \qquad \mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$$

$$\mathbf{u}_{k+1} = \mathbf{a}_{k+1} - \sum_{i=1}^{k} (\mathbf{a}_{k+1} \cdot \mathbf{e}_i)\mathbf{e}_i, \qquad \mathbf{e}_{k+1} = \frac{\mathbf{u}_{k+1}}{\|\mathbf{u}_{k+1}\|}$$

Note that $\|\cdot\|$ is the L_2 norm.

QR Factorization

The resulting QR factorization is given by $A = QR$, where

$$A = \begin{bmatrix} \mathbf{a}_1 | & \mathbf{a}_2 & | & \dots & | & \mathbf{a}_n \end{bmatrix}$$

$$Q = \begin{bmatrix} \mathbf{e}_1 | & \mathbf{e}_2 & | & \dots & | & \mathbf{e}_n \end{bmatrix}$$

$$R = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{e}_1 & \mathbf{a}_2 \cdot \mathbf{e}_1 & \dots & \mathbf{a}_n \cdot \mathbf{e}_1 \\ 0 & \mathbf{a}_2 \cdot \mathbf{e}_2 & \dots & \mathbf{a}_n \cdot \mathbf{e}_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{a}_n \cdot \mathbf{e}_n \end{bmatrix}$$

Once we find $\mathbf{e}_1, \dots, \mathbf{e}_n$, it is straightforward to construct the Q and R matrices.

Example: Initial Matrix and Vectors

Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

with the column vectors:

$$\mathbf{a}_1 = (1, 1, 0)^T$$
$$\mathbf{a}_2 = (1, 0, 1)^T$$
$$\mathbf{a}_3 = (0, 1, 1)^T$$

Note: From now on, T notation for column vectors will be dropped for simplicity, but all vectors should be understood as column vectors.

Example: Step 1 – Finding $\mathbf{u}_1$ and $\mathbf{e}_1$

Performing the Gram-Schmidt procedure:

$$\begin{aligned}
\mathbf{u}_1 &= \mathbf{a}_1 = (1, 1, 0) \\
\|\mathbf{u}_1\| &= \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2} \\
\mathbf{e}_1 &= \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \frac{1}{\sqrt{2}}(1, 1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right).
\end{aligned}$$

Example: Step 2 – Finding $\mathbf{u}_2$ and $\mathbf{e}_2$

$$\mathbf{u}_2 = \mathbf{a}_2 - (\mathbf{a}_2 \cdot \mathbf{e}_1)\mathbf{e}_1$$

$$\mathbf{a}_2 \cdot \mathbf{e}_1 = (1,0,1) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) = \frac{1}{\sqrt{2}}$$

$$\mathbf{u}_2 = (1,0,1) - \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

$$= (1,0,1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

$$\|\mathbf{u}_2\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + 1^2} = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$$

$$\mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{3/2}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right).$$

Example: Step 3 – Finding $\mathbf{u}_3$ (Part 1)

$$\mathbf{u}_3 = \mathbf{a}_3 - (\mathbf{a}_3 \cdot \mathbf{e}_1)\mathbf{e}_1 - (\mathbf{a}_3 \cdot \mathbf{e}_2)\mathbf{e}_2$$

$$\mathbf{a}_3 \cdot \mathbf{e}_1 = (0,1,1) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) = \frac{1}{\sqrt{2}}$$

$$(\mathbf{a}_3 \cdot \mathbf{e}_1)\mathbf{e}_1 = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Now for the second term:

$$\begin{aligned}\mathbf{a}_3 \cdot \mathbf{e}_2 &= (0,1,1) \cdot \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right) \\ &= 0 \cdot \frac{1}{\sqrt{6}} + 1 \cdot \left(-\frac{1}{\sqrt{6}}\right) + 1 \cdot \frac{2}{\sqrt{6}} \\ &= -\frac{1}{\sqrt{6}} + \frac{2}{\sqrt{6}} = \frac{1}{\sqrt{6}}\end{aligned}$$

Example: Step 3 – Finding $\mathbf{u}_3$ and $\mathbf{e}_3$ (Part 2)

Continuing from the previous slide:

$$(\mathbf{a}_3 \cdot \mathbf{e}_2)\mathbf{e}_2 = \frac{1}{\sqrt{6}}\left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right) = \left(\frac{1}{6}, -\frac{1}{6}, \frac{2}{6}\right)$$

$$\begin{aligned}
\mathbf{u}_3 &= (0,1,1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{2}{6}\right) \\
&= \left(0 - \frac{1}{2} - \frac{1}{6}, 1 - \frac{1}{2} - \left(-\frac{1}{6}\right), 1 - 0 - \frac{2}{6}\right) \\
&= \left(-\frac{3+1}{6}, \frac{3+1}{6}, \frac{6-2}{6}\right) = \left(-\frac{4}{6}, \frac{4}{6}, \frac{4}{6}\right) = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)
\end{aligned}$$

$$\|\mathbf{u}_3\| = \sqrt{\left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{12}{9}} = \frac{2}{\sqrt{3}}$$

$$\mathbf{e}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \frac{1}{2/\sqrt{3}}\left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right) = \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right).$$

Example: Final Q and R Matrices

Thus, the matrices Q and R are:

$$Q = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$R = \begin{bmatrix} \mathbf{a}_1 \cdot \mathbf{e}_1 & \mathbf{a}_2 \cdot \mathbf{e}_1 & \mathbf{a}_3 \cdot \mathbf{e}_1 \\ 0 & \mathbf{a}_2 \cdot \mathbf{e}_2 & \mathbf{a}_3 \cdot \mathbf{e}_2 \\ 0 & 0 & \mathbf{a}_3 \cdot \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{\sqrt{3}}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$